

Non-Uniqueness of Distribution-Shift Attribution, and Why We Localize Subsets Instead

Technical note

Abstract

For a marketplace comparing an existing batch of data against a new batch, we show a basic *impossibility*: the *total* distribution shift is identifiable and order-invariant, but the *per-feature contribution* to it is not uniquely defined — it depends on the order in which features are considered. We demonstrate this empirically with random feature-addition paths under two metrics: Maximum Mean Discrepancy (MMD) for covariate shift $P(X)$ and the pseudo-outcome / R -risk for concept drift $P(Y | X)$. The consequence is methodological: at any level of the entity hierarchy one should *localize a responsible subset* (well-posed) rather than attribute per-feature proportions (ill-posed). We give the construction, the empirical result, the resulting multi-level drill-down, a worked business example, and how such a system should be evaluated.

1 Setup

Orders (observations) carry raw attributes and are labelled by batch $W \in \{0, 1\}$ (existing vs. new). We roll orders up to an entity axis $\mathbf{e}$ (merchant, buyer, item, region, ...) with a feature map Φ producing an entity–feature matrix $x \in \mathbb{R}^p$ (rich + rolling aggregations), with features grouped by raw attribute, $g : \{1, \dots, p\} \rightarrow \mathcal{G}$. Let P_0, P_1 be the two-batch feature distributions. A characteristic-kernel MMD measures covariate shift, $\text{MMD}^2(P_0, P_1) = 0 \iff P_0 = P_1$; with batch as “treatment”, the R -risk of the induced pseudo-outcome regression measures concept drift and is insensitive to pure covariate shift.

2 Path-order attribution and its non-uniqueness

Let $T(S)$ be a shift metric (MMD² or R -risk) evaluated on the feature subset $S \subseteq \{1, \dots, p\}$. Given an ordering π of the features, the *addition path* is the curve $S_0 = \emptyset \subset S_1 \subset \dots \subset S_p$ with $S_k = \{\pi_1, \dots, \pi_k\}$, and the marginal credited to feature π_k is $T(S_k) - T(S_{k-1})$. Sampling many random π and averaging the credits yields the Shapley value; but the credits along any single path are what a “how much did feature f contribute” question actually asks.

Proposition 1 (Total identifiable, path non-unique). *The endpoint $T(S_p) = T(\{1, \dots, p\})$ is invariant to the order π : the total shift is uniquely defined and estimable. The intermediate values $T(S_k)$, $0 < k < p$, and hence the per-feature marginals, are not invariant to π whenever the features are dependent; there is no order-free “contribution of feature f ”. Shapley restores uniqueness only by averaging over all orders, i.e. by adopting a convention, not by identifying a truth.*

Empirical demonstration. We draw random feature-addition paths on the repository’s banded-Toeplitz covariate-shift and concept-drift DGMs and record the metric at every step. Figure 1 shows the result: all paths converge to the same full-set total (red, order-invariant), but the accrual curves

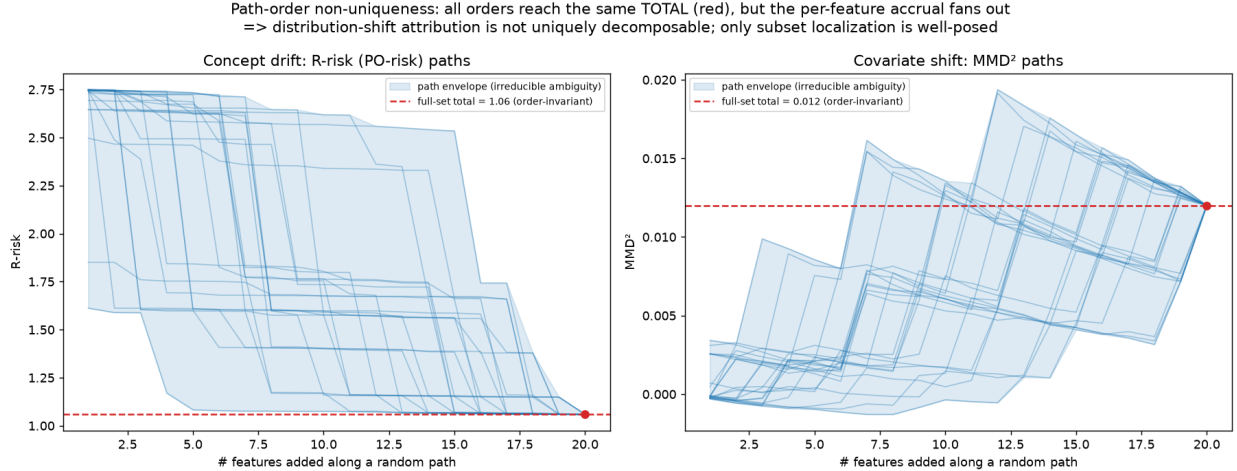


Figure 1: Random feature-addition paths. Every order reaches the same total (red, order-invariant); the per-feature accrual fans out (shaded envelope = irreducible ambiguity). Left: concept drift (R -risk); right: covariate shift (MMD^2).

fan out in between; the shaded band is the *irreducible attribution ambiguity*. For concept drift the maximum gap between two orders is $\approx$ the entire total (~ 1.0 against a total of ~ 1.06): two feature orderings can disagree about “how much has been explained so far” by essentially the whole shift. The per-step standard deviation across orders is 30–40% of the total for both metrics.

Remark 1 (Any number of levels). *The argument is scale-free: it applies to features within one entity, and equally to the coarser “how much does each child level contribute” question. Hence a distribution shift is not uniquely decomposable at any level of the hierarchy.*

3 The well-posed object: multi-level subset localization

While “how much” is not identifiable, “which subset carries the shift” is: the total is order-invariant, and the selection of the responsible group is stable across attribution conventions and across resamples. We therefore localize recursively, and never report proportions. The selection deliberately uses a single global detection plus a leave-one-group-out ablation and a stability check — *no multiple-testing machinery*, which keeps the procedure simple and easy to trust:

1. **Detect** the shift globally with one MMD / R -risk permutation test (order-invariant total).
2. **Localize (level 1)** to the responsible attribute-group subset by *leave-one-group-out (LOGO) ablation* — the groups whose removal collapses the shift — kept only if stable across bootstrap resamples.
3. **Drill (level 2)** by *restricting the order table to the entities selected at level 1* (a conditional drill-down: keep only the flagged subset’s rows), then re-running the same selection on the next aggregation key (e.g. item / product vertical) *within that subset* — again returning a subset, not a share. Level 2 is therefore always evaluated on the level-1 subset, not the whole population.

4. **Close the loop:** each selected subset carries an owner, a candidate rule (a KS-optimal split threshold), a decision (monitor / reweight / retrain / collect labels), and a trend status computed by diffing against the previous run.

4 Why LOGO, why these modalities, and why the subset is the unit

Why LOGO. We rank and select groups by the leave-one-group-out ablation $\phi_g = T(\text{all}) - T(\text{all} \setminus g)$, the drop in the shift metric when a group’s features are removed. It is the right primitive here for three reasons. (i) It answers the operational question directly — “if we ignore or repair this group, does the detected shift disappear?” — which is how an owner reasons and what makes the output actionable. (ii) It is a *conditional* importance (given all other features present), so a group is credited only for what the rest does not already explain: it flags *non-redundant* drivers rather than everything merely correlated with the shift. (iii) It is cheap ($K+1$ metric evaluations), its sign is meaningful (removing a true driver lowers the metric; removing noise does not), and it needs no attribution convention. Applied at the raw-attribute *group* level it is immune to the within-group correlation that would otherwise dilute every single aggregation to near zero.

Why the subset — not the feature — is the effective unit. By the non-uniqueness result the total is order-invariant but the per-feature split is not; the object that *is* invariant and estimable is the contribution of a whole subset (include or exclude the group as a unit). Empirically the group-level answer is also stable across attribution conventions (standalone, LOGO, and Shapley agree on *which* groups) and across resamples, while the per-feature answer is not. The grouping is not arbitrary: a real shift typically enters through a raw signal — a collection/definition change in GMV, a pricing change, a new buyer cohort — which moves *all* of that signal’s aggregations together, so the raw-attribute group is the natural causal and operational unit. Selecting that unit is therefore both statistically well-posed (invariant, stable) and operationally meaningful (it maps to an owner, a rule, and an action).

Why these modalities (MMD and PO-/R-risk). A characteristic-kernel MMD is zero iff the two batches share the same $P(X)$, and its LOGO drop is a genuine reduction in a distributional distance that is sensitive to shifts in any moment (mean, scale, tails), so it does not miss non-linear covariate shifts. The PO-/R-risk targets the orthogonal, outcome-relevant question — has $P(Y | X)$ changed — and is insensitive to pure covariate shift, so the two modalities cover the two failure modes without double-counting. Both are permutation-exact and assumption-light, which is what makes a LOGO drop built on them a trustworthy, convention-free number — in contrast to Shapley, which becomes unique only by averaging over an arbitrary set of orders.

5 A worked business example

Symptom. In the new week’s batch, the global test fires. Level-1 localization on the *region* axis selects the activity subset — concretely the order-count feature `n_orders` (and order-value aggregations) shifted *up* for a subset of users in region *R*: those buyers are placing more (and larger) orders than the existing batch.

Non-uniqueness, honestly. We do *not* claim “region R ’s users account for 43% of the shift” — by the proposition that number is order/convention dependent. We claim the shift is *localized to region R ’s user subset*, with a stability of, say, 0.98 and a split rule `n_orders per user > t`.

Drill one level — *inside the flagged user subset*. We first restrict the order table to exactly the level-1 subset (the flagged region- R users, `n_orders per user > t`), and only then aggregate those users’ orders on the *item / product-vertical* axis and re-run the two-stage selection. This conditional drill localizes the extra orders to a specific **product vertical** (item category C) *within that user subset*: among the flagged region- R users the surge concentrates in category C (e.g. “fresh grocery”), not uniformly across the catalogue. The localization is thus **product-vertical** $\times$ **user-subset** (category C conditioned on the region- R user segment), output as subset = {region R users, category C } with its own split rule and stability. Evaluating level 2 on the whole population instead of on the level-1 subset would confound the two levels and is explicitly avoided.

Action / closed loop. The subset routes to the category- C / region- R owner; the candidate rule becomes a monitor; the decision (e.g. *reweight* if the mapping is unchanged, *collect labels / retrain* if the outcome relationship moved) is attached; the **trend** field marks it *resolved* once the subset stops flagging. The business learns *where* and *who* and *what to do* — never a spurious percentage.

Remark 2 (Why this is the right granularity). “43% from category C ” would be *unfalsifiable (non-identifiable) and un-actionable*. “The new-batch order surge is localized to category C among region- R users; here is the threshold and the owner” is *checkable, routable, and monitorable* — it closes the loop.

Results of the two-level procedure. We inject a GMV covariate shift concentrated in a *single product item category* (Fresh Grocery) for new merchants (200 merchants, 6 categories, 24k orders) and run the procedure. Level 1 (merchant axis) selects the **gmV** attribute group by LOGO ablation with full stability; the conditional Level 2 then drills into the **product item category** and isolates **Fresh Grocery** exactly, while every other category is non-significant (Table 1). No proportions are reported — only the selected subsets, their ablation/gap magnitude, stability, and permutation p .

6 How to evaluate such a system

Because per-feature proportions are not identifiable, they must not be the yard stick. Evaluate only identifiable targets:

Target	Identifiable?	Metric
Total / detection	yes (order-invariant)	test power; type-I under permuted labels
Subset localization	yes (convention-stable)	subset recall / precision vs. ground truth
Stability	yes	bootstrap selection frequency
Decision layer	yes (with labels)	downstream Δ loss backtest
Per-feature proportion	no	— (do not evaluate)

The principle: judge a method by the questions it claims to answer (detection, subset localization, stability, downstream impact), not by a question that is provably unanswerable. The non-identifiability itself is demonstrated empirically (Figure 1), not hidden.

Level (unit)	Candidate	LOGO / gap	Stability	Perm. p
<i>Level 1 — merchant axis</i> (global MMD ² = 0.281, p = 0.003)				
	gmv (selected)	+0.283	1.00	—
	quantity	−0.277	0.85	—
<i>Level 2 — product item category</i> (conditional on gmv, new vs. existing)				
	Fresh Grocery (c^* , selected)	55.88	—	0.002
	Home & Kitchen	0.77	—	0.158
	Apparel	0.58	—	0.240
	Electronics	0.29	—	0.621
	Beauty	0.26	—	0.583
	Toys	0.03	—	0.932

Table 1: Two-level conditional subset localization (`fsds_cross_level.py`). Level 1 selects the **gmv** attribute subset (LOGO ablation + bootstrap stability); the conditional Level 2 drills into the **product item category** and selects exactly the shifted one, **Fresh Grocery** (gap = new−existing mean GMV within the category; merchant-block permutation p). The order surge is localized to a single product item category, not spread as proportions; ground truth is recovered at both levels.

7 Conclusion

The total distribution shift is identifiable; its per-feature decomposition is not. This is not a limitation of a particular estimator but a property of dependent features under any additive path attribution. The productive response is to *localize responsible subsets* recursively down the entity hierarchy and attach owner / rule / decision / trend to each — turning an unanswerable “how much” into an operational “which subset, who, and what to do”.